Definition 1. For any binary string π of length T, a *cylindric plane partition* with profile π may be defined as a sequence of integer partitions:

$$(\mu^0, \mu^1, \dots \mu^T) \qquad \mu^0 = \mu^T \tag{1}$$

such that if $\pi_k = 1$ then μ^k/μ^{k-1} is a *horizontal strip*, otherwise if $\pi_k = 0$ then μ^{k-1}/μ^k is a horizontal strip.

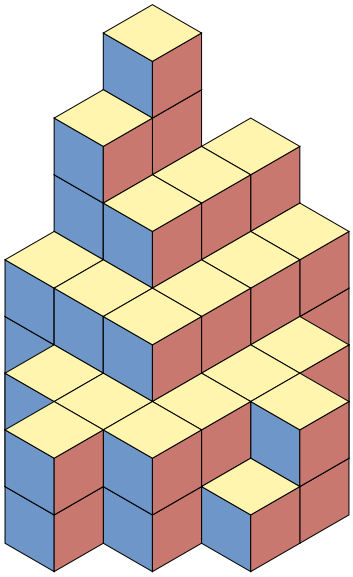
There are several different ways to represent a cylindric plan partition visually. For example, consider the cylindric plane partition:

$$\mathfrak{c} = \big((3,2,2),\ (4,3,2,1),\ (4,3,2),\ (6,4,3,2),\ (5,3,2),\ (3,2,2)\big)$$

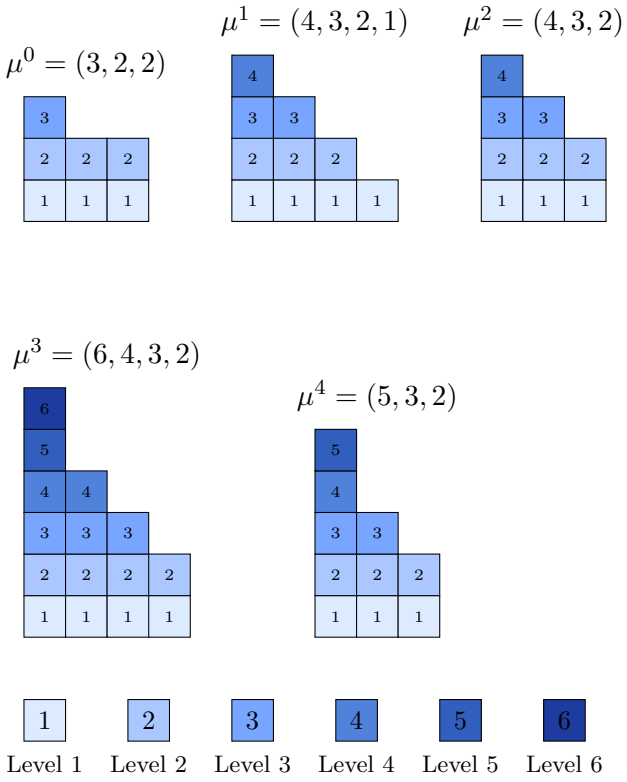
with profile $\pi = 10100$. This can be represented as a "semi-standard cylindric tableau". Each "cell" has a "label". The labels are weakly decreasing along both rows and columns. The sequence of integer partitions is read off the diagonals (from right to left).

	4	3		
6	4	3	2	
5	4	3	2	2
3	3	3	2	1
	2	2	2	0
		2	0	0

For a "cell" with label k we can imagine k "cubes" stacked up on top of the page. Each "cube" corresponds to a "square" in one of the underlying integer partitions.



Each "square" in column k of a Young diagram belongs to **level** k. These are the "cubes" of height k. We **crush** level k by replacing all labels k with $k-1$.



Example: crushing ring 3. Six cells in the labelled diagram carry label 3; all become 2. The resulting partition sequence is:

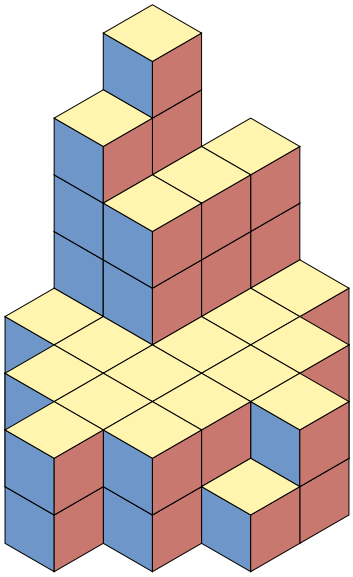
$$\mu^0 = (2,2,2), \quad \mu^1 = (4,2,2,1), \quad \mu^2 = (4,2,2), \quad \mu^3 = (6,4,2,2), \quad \mu^4 = (5,2,2).$$

	Before	**After**	**Cells changed**
μ^0	$(3,2,2)$	$(2,2,2)$	1
μ^1	$(4,3,2,1)$	$(4,2,2,1)$	1
μ^2	$(4,3,2)$	$(4,2,2)$	1
μ^3	$(6,4,3,2)$	$(6,4,2,2)$	1
μ^4	$(5,3,2)$	$(5,2,2)$	1

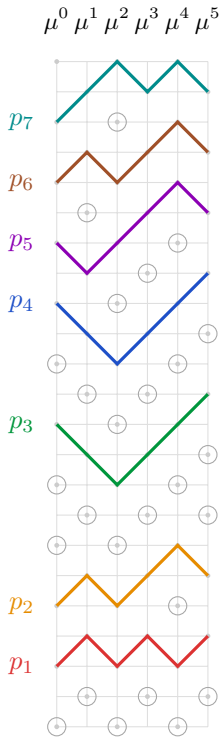
Labelled diagram after crushing level 3:

	4	2		
6	4	2	2	
5	4	2	2	2
2	2	2	2	1
	2	2	2	0
		2	0	0

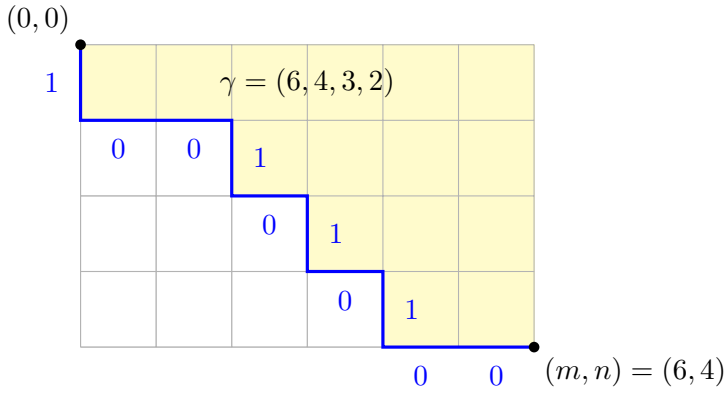
3D isometric view after crushing level 3:



Another way to represent cylindric plane partitions is with non-intersecting lattice paths on a cylinder.



This construction uses the natural bijection between integer partitions and binary strings. The profile $\pi = 1001010100$ encodes the partition $\gamma = (6, 4, 3, 2)$ via the boundary path of the Young diagram inside a 6×4 rectangle. Reading π left to right: 1 = down step, 0 = horizontal step.

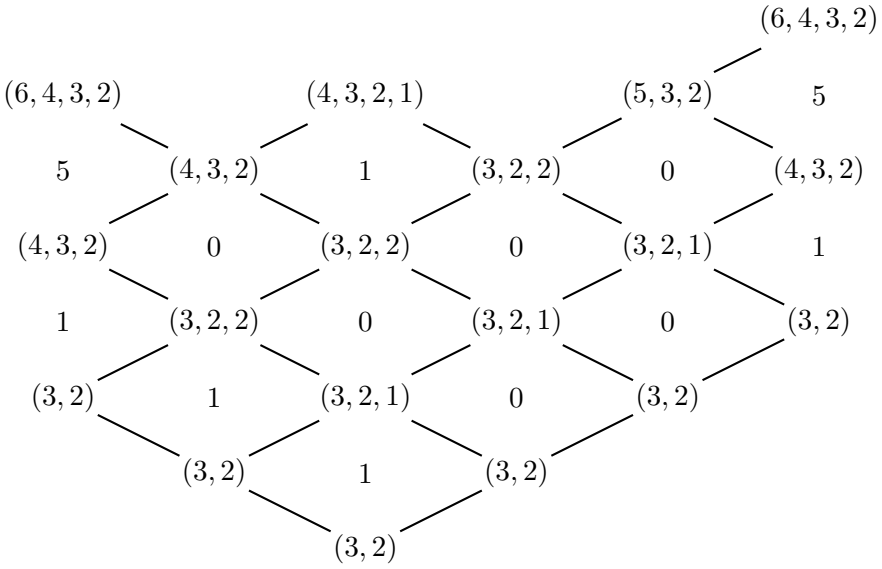


Crushing level k of a family of non-intersecting lattice paths is equivalent to pushing the kth path flush up against the one above it.

The formula for the enumeration of Cylindric Plane Partitions was originally given by Borodin:

$$\sum_{\mathfrak{c}\in \mathrm{CPP}(\pi)} z^{|\mathfrak{c}|} = \prod_{n\geq 0} \left(\frac{1}{1-z^{(n+1)T}} \prod_{\substack{i<j \\ \pi_i > \pi_j}} \frac{1}{1-z^{j-i+nT}} \prod_{\substack{i>j \\ \pi_i > \pi_j}} \frac{1}{1-z^{j-i+(n+1)T}} \right) \tag{2}$$

In my thesis I proved this result using growth diagrams:



The right hand side of the equation can be interpreted as a pair $(\gamma, \mathfrak{d})$ where γ is an integer partition and $\mathfrak{d}$ is a "hook length weighted labelled cylindric diagram". The proof makes use of "local rules". To understand the local rules we need to look at Cauchy's Identity.

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Proposition 2. *If X_n is the alphabet $X_n = x_1 + x_2 + \cdots + x_n$ and Y_n is the alphabet $Y_n = y_1 + y_2 + \cdots + y_n$ then:*

$$\sum_{\lambda} S_\lambda[X_n]S_\lambda[Y_n] = \prod_{i,j=1}^{n} \frac{1}{1-x_i y_j}$$

Proof. The case $n = 1$ is trivial. We assume that the result is true for all $k < n$. By the Pieri formula and the induction assumption, we have on the right hand side:

$$\begin{aligned} RHS(n) &= \left(\sum_r h_r[X_{n-1}]x_n^r\right)\left(\sum_s h_s[Y_{n-1}]y_n^s\right)\left(\sum_\mu S_\mu[X_{n-1}]S_\mu[Y_{n-1}]\right)\frac{1}{1-x_n y_n} \\ &= \sum_{\alpha,\beta} S_\alpha[X_{n-1}]S_\beta[Y_{n-1}]\left(\frac{1}{1-x_n y_n}\sum_{\mu\in D(\alpha)\cap D(\beta)} x_n^{|\alpha/\mu|}y_n^{|\beta/\mu|}\right) \end{aligned}$$

while by the dual Pieri formula we have on the left hand side:

$$\begin{aligned} LHS(n) &= \sum_\lambda S_\lambda[X_{n-1}+x_n]S_\lambda[X_{n-1}+y_n] \\ &= \sum_{\alpha,\beta} S_\alpha[X_{n-1}]S_\beta[Y_{n-1}]\left(\sum_{\lambda\in U(\alpha)\cap U(\beta)} x_n^{|\lambda/\alpha|}y_n^{|\lambda/\beta|}\right) \end{aligned}$$

In order to complete this proof, we need, for any given α and β a bijection between the set of pairs (m,μ) with $\mu \in D(\alpha)\cup D(\beta)$ and m a non-negative integer and the set of partitions $\lambda \in U(\alpha)\cup U(\beta)$ in such a way that:

$$|\lambda| = |\alpha| + |\beta| - |\mu| + m$$

A *local rule* is just such a bijection.

$$\mathfrak{U}_{\alpha,\beta} : (\mathbb{Z}_{\geq 0}, D(\alpha)\cup D(\beta)) \to U(\alpha)\cup U(\beta) \tag{3}$$

□

Definition 3. Consider the graded generating series for cylindric play partitions with profile c:

$$\mathrm{GK}_c(z,q) = \sum_m g_m(q) z^m = \sum_{\pi \in \mathrm{CPP}(c)} q^{|\pi|} z^{\max(\pi)}. \tag{4}$$

The *Q-polynomial* is defined by:

$$Q_{n,c}(q) = (q;q)_n \cdot [z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big). \tag{5}$$

Conjecture 4 (Warnaar–Corteel)**.** *For all $n \geq 0$ and all rank-3 profiles c,*

$$Q_{n,c}(q) \in \mathbb{N}[q].$$

Proposition 5. *If $g_m(q) = \sum_n \mathcal{G}_{n,c} q^n$ then with some straightforeward manipulation we can show that:*

$$\widetilde{\mathcal{G}}_{n,c} = \bigsqcup_{j=0}^{n} \left\{ (S,U,\nu) : \begin{array}{l} S \subseteq \{1,\dots,n\},\ |S| = j, \\ U \subseteq \{1,\dots,n-j\}, \\ \nu \in \mathrm{CPP}(c),\ \max(\nu) = n-j \end{array} \right\}, \tag{6}$$

with sign $\varepsilon(S,U,\nu) = (-1)^{|S|+|U|}$ and weight $w(S,U,\nu) = \sigma(S) + \sigma(U) + |\nu|$.

Proof. We expand the three factors in (5).

Step 1: Euler's identity. The q-Pochhammer product has the expansion

$$(zq;q)_\infty = \sum_{j \geq 0} \frac{(-1)^j\, q^{j(j+1)/2}}{(q;q)_j}\, z^j.$$

Step 2: Extract the z^n coefficient. Multiplying by $\mathrm{GK}_c(z,q) = \sum_m g_m(q)\, z^m$ and extracting $[z^n]$:

$$[z^n]\big((zq;q)_\infty \cdot \mathrm{GK}_c(z,q)\big) = \sum_{j=0}^{n} \frac{(-1)^j\, q^{j(j+1)/2}}{(q;q)_j}\, g_{n-j}(q).$$

Multiplying by $(q;q)_n$ and using $\frac{(q;q)_n}{(q;q)_j} = \begin{bmatrix} n \\ j \end{bmatrix} (q;q)_{n-j}$:

$$Q_{n,c}(q) = \sum_{j=0}^{n} (-1)^j\, q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} (q;q)_{n-j}\, g_{n-j}(q). \tag{7}$$

Step 3: Combinatorial interpretation of each factor.

Recall that the *q-binomial coefficient* counts partitions fitting inside a $j \times (n-j)$ rectangle:

$$\begin{bmatrix} n \\ j \end{bmatrix} := \frac{(q;q)_n}{(q;q)_j\,(q;q)_{n-j}} = \sum_{\lambda \subseteq (n-j)^j} q^{|\lambda|}.$$

One can show that for integers $0 \leq j \leq n$,

$$q^{j(j+1)/2} \begin{bmatrix} n \\ j \end{bmatrix} = \sum_{\substack{S \subseteq \{1,\dots,n\} \\ |S| = j}} q^{\sigma(S)}, \tag{8}$$

where $\sigma(S) = \sum_{s \in S} s$. Similarly, the finite q-Pochhammer symbol expands as:

$$(q;q)_m = \prod_{k=1}^{m} (1 - q^k) = \sum_{U \subseteq \{1,\dots,m\}} (-1)^{|U|}\, q^{\sigma(U)}. \tag{9}$$

Finally, $g_m(q) = \sum_{\nu \in \mathrm{CPP}(c),\, \max(\nu) = m} q^{|\nu|}$ by definition of GK_c.

Step 4: Combine. Substituting (8), (9), and the definition of g_m into (7) with $m = n - j$:

$$Q_{n,c}(q) = \sum_{j=0}^{n} \sum_{\substack{S \subseteq \{1,\dots,n\} \\ |S| = j}} \sum_{U \subseteq \{1,\dots,n-j\}} \sum_{\substack{\nu \in \mathrm{CPP}(c) \\ \max(\nu) = n-j}} (-1)^{|S|+|U|}\, q^{\sigma(S)+\sigma(U)+|\nu|}.$$

Since $j = |S|$, this is precisely the signed generating function over the ground set $\widetilde{\mathcal{G}}_{n,c}$. □

1 Multi-variate analog

We replace q^k with q_k and when c is a cylindric plane partition we replace q^c with $q_{\lambda_1} q_{\lambda_2} q_{\lambda_3}$. The definition of $Q_n(q_1, q_2, ...)$ becomes:

$$Q_n(q_1, q_2....) = \sum_k (-1)^k e_k(q_1, q_2, ...)[z^n] \sum_k (-1)^k e_k(q_1, q_2, ...) \sum_c q_{\lambda_1} q_{\lambda_2} q_{\lambda_3} z^{max(c)} \quad (10)$$

$$= \quad (11)$$

Does the right hand side look like this:

$$\sum (S, U, \nu)(\prod_{s \in S} \prod_{u \in u} q_s q_u) \sum q_{\lambda_1} q_{\lambda_2} q_{\lambda_3}$$

]

Start with a cylindric plane partition λ with maximum entry n. This corresponds to a family of $n+1$ non-intersecting lattice paths. The set S is of size j. So we remove j paths (not including the top one), corresponding to the elements of S. This leaves a partition γ with maximum entry $n-j$. Now we have U. This set contains $n-j$ elements. Of the remaining paths (excluding the top one) we "crush" those corresponding to the elements of U. That is we push them up flush against the path immediately above. This gives us a new partition mu also with maximum part $n-j$. This process cannot be revursed. It defines an "equivalence relation" on the set of all cylindric plane partitions with maximum part n.